3.4

Q2 a. $Pr(X=2) = Pr(X=2, Y=0) + \cdots + Pr(X=2, Y=4)$
$= 0.27$

b. $Pr(Y \geq 2) = 1 - Pr(Y=0) - Pr(Y=1)$
$= 0.53$

c. $Pr(X<2 \,\&\, Y \leq 2) = 0.49$

d $Pr(X=Y) = 0.3$

e. $Pr(X>Y) = 0.25$

Q4. a. the value of the constant c

$\int_0^2 \int_0^1 cy^2 dy dx = 1 \Rightarrow c\int_0^1 \left[\frac{y^3}{3}\right]_0^1 dx = \frac{2c}{3}$

$\therefore c = \frac{3}{2}$

b. $Pr(X+Y>2) \Rightarrow \int_0^2 \int_0^1 f(x,y) dy$ in $x+y>2$.

$\therefore Pr(X+Y>2) = \int_0^1 \int_{2-x}^1 \frac{3}{2} y^2 dy dx + \int_1^2 \int_0^1 \frac{3}{2} y^2 dy dx = \frac{3}{4}$

试用水印

c. $Pr(Y<\frac{1}{2}) = \int_0^2 \int_0^{\frac{1}{2}} \frac{3}{2} y^2 dy dx = \frac{1}{8}$

d. $Pr(X \leq 1) = \int_0^1 \int_0^1 \frac{3}{2} y^2 dy dx = \frac{1}{2}$

e. $Pr(X=3Y)$ is in $\{x \in [0,2]$ $y \in (0, \frac{2}{3})$ is ___

Q8. $F(x,y) = \frac{1}{156} xy(x^2+y)$

a. $Pr(1 \leq X \leq 2 \,\&\, 1 < Y \leq 2) = F(2,2) + F(1,1) - F(2,1) - F(1,2)$
$= \frac{7}{156}$

b. $Pr(2 \leq X \leq 4 \,\&\, 2 < Y \leq 4) = 0.$

c. c.d.f of Y

$F_Y(y) = \int_0^3 F(x,y) dx = \frac{y}{156} \int_0^3 (x^3 + xy) dx$
$= \frac{3y}{312}(27 + 18y)$

d. p.d.f of X, Y

$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$

3.5

Q3. $f(x,y) = \begin{cases} \frac{3}{2}y^2 & \text{for } 0 \le x \le 2 \text{ and } 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$

a. to find p.d.f of X.Y.

$f_X(x) = \int_0^1 \frac{3}{2}y^2 dy = \frac{1}{2}$ for $0 \le x \le 2$

$f_Y(y) = \int_0^2 \frac{3}{2}y^2 dx = 3y^2$ for $0 \le y \le 1$

b. if X & Y independent

then. $f(x,y) = f_X(x) \cdot f_Y(y)$

$f(x,y) = \frac{3}{2}y^2 = \frac{1}{2} \cdot 3y^2$

$\therefore$ q.e.d. X & Y independent

c if independent $P(X<1 \,\&\, Y \ge \frac{1}{2}) = P(X<1) \cdot P(Y \ge \frac{1}{2})$

$P(X<1) = \frac{1}{2}$ $P(Y \ge \frac{1}{2}) = \frac{7}{8}$

$P(X<1 \,\&\, Y \ge \frac{1}{2}) = \int_0^1 \frac{21}{16} dx \Leftrightarrow P = 1$

$P(X<1) \cdot P(Y \ge \frac{1}{2}) = \frac{7}{16} < 1$

$\therefore$ is not independent.

Q9. $S = \{(x,y) : 0 \le x \le 2 \,\&\, 1 \le y \le 4\}$

a. joint p.d.f. $f(x,y) = \begin{cases} \frac{1}{6} & 0 \le x \le 2 \,\&\, 1 \le y \le 4 \\ 0 & \text{otherwise} \end{cases}$

marginal p.d.f. of X is

$f_X(x) = \begin{cases} \int_1^4 f(x,y)dy = \int_1^4 \frac{1}{6} dy = \frac{1}{2} & 0 \le x \le 2 \\ 0 & \text{otherwise.} \end{cases}$

marginal p.d.f. of Y

$f_Y(y) = \begin{cases} \int_0^2 f(x,y)dx = \frac{1}{3} & 1 \le y \le 4 \\ 0 & \text{otherwise.} \end{cases}$

b. if X & Y independent

$f(x,y) = f_X(x) \cdot f_Y(y)$

$\because \frac{1}{6} = \frac{1}{3} \times \frac{1}{2}$ in $0 \le x \le 2$, $1 \le y \le 4$

$\therefore$ X & Y independent

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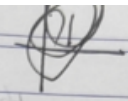
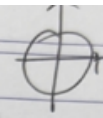
Q.10 $\because S=\{(x,y): x^2+y^2\le 1\}$





a. joint p.d.f. of X & Y $\quad f_{XY}(x,y)=\frac{1}{\pi}$. $(x,y)\in S$

$f_X(x)=\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\frac{1}{\pi}\,dy=\frac{2}{\pi}\sqrt{1-x^2}\quad -1\le x\le 1$

$f_Y(y)=\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}}\frac{1}{\pi}\,dx=\frac{2}{\pi}\sqrt{1-y^2}\quad -1\le y\le 1$

b. $f_{XY}(x,y)=\frac{1}{\pi}$ but $f_X(x)\cdot f_Y(y)\ne\frac{1}{\pi}$

$\therefore X$ & Y not independent

Q6. a. joint p.d.f of X & Y

$f_{X,Y}(x,y)=g(x)\cdot g(y)=\frac{9}{64}x^2y^2\quad 0\le x\le 2$ & $0\le y\le 2$

b. $Pr(X=Y)=0$.

c. $Pr(X>Y)=\frac{1}{2}$

d $Pr(X+Y\le 1)=\int_0^1\int_0^{1-x}\frac{9}{64}x^2y^2\,dy\,dx=\frac{3}{64}$

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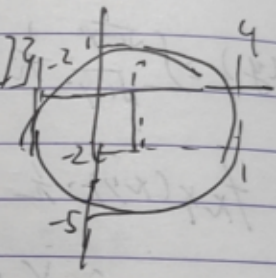
3.6

Q2. a. $P=\frac{0.09}{0.33}=0.1212$

b. $P(MTO)=0.27$

$P(MTO\mid S)=\frac{0.1}{0.27}=0.3704$

Q3. $S=\{(x,y): (x-1)^2+(y+2)^2\le 9\}$



a. joint p.d.f

$f_{X,Y}(x,y)=\frac{1}{S}=\frac{1}{9\pi}$

$f_X(x)=\int_{-\sqrt{9-(x-1)^2}-2}^{\sqrt{9-(x-1)^2}-2}f_{XY}(x,y)\,dy$

$\Rightarrow f_X(x)=\frac{2\sqrt{9-(x-1)^2}}{9\pi}$

$\therefore f_{Y|X}(y|x)=\frac{f_{XY}(x,y)}{f_X(x)}=\frac{1}{2\sqrt{9-(x-1)^2}}\quad -\sqrt{9-(x-1)^2}-2\le y\le\sqrt{9-(x-1)^2}-2$

$Pr(Y>0\mid X=2)$ when $x=2$ $\quad -2\sqrt{2}-2\le y\le 2\sqrt{2}-2$

$f_{Y|X}(y|2)=\frac{1}{4\sqrt{2}}$

$Pr(Y>0\mid X=2)=\int_0^{2\sqrt{2}-2}f_{Y|X}(y|2)\,dy=\frac{2-\sqrt{2}}{4}$

显示工具菜单 Ctrl +Alt+左键 选择并
显示模式选择 Ctrl +L 选择相连

Q4 a. p.d.f of X. for.

$$f_X(x) = 2\int_0^1 c(x+y^2)\,dy = c(x+\frac{1}{3}) = cx+\frac{c}{3}$$

b. $P_r(X<1/2 \mid Y=1/2)$

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